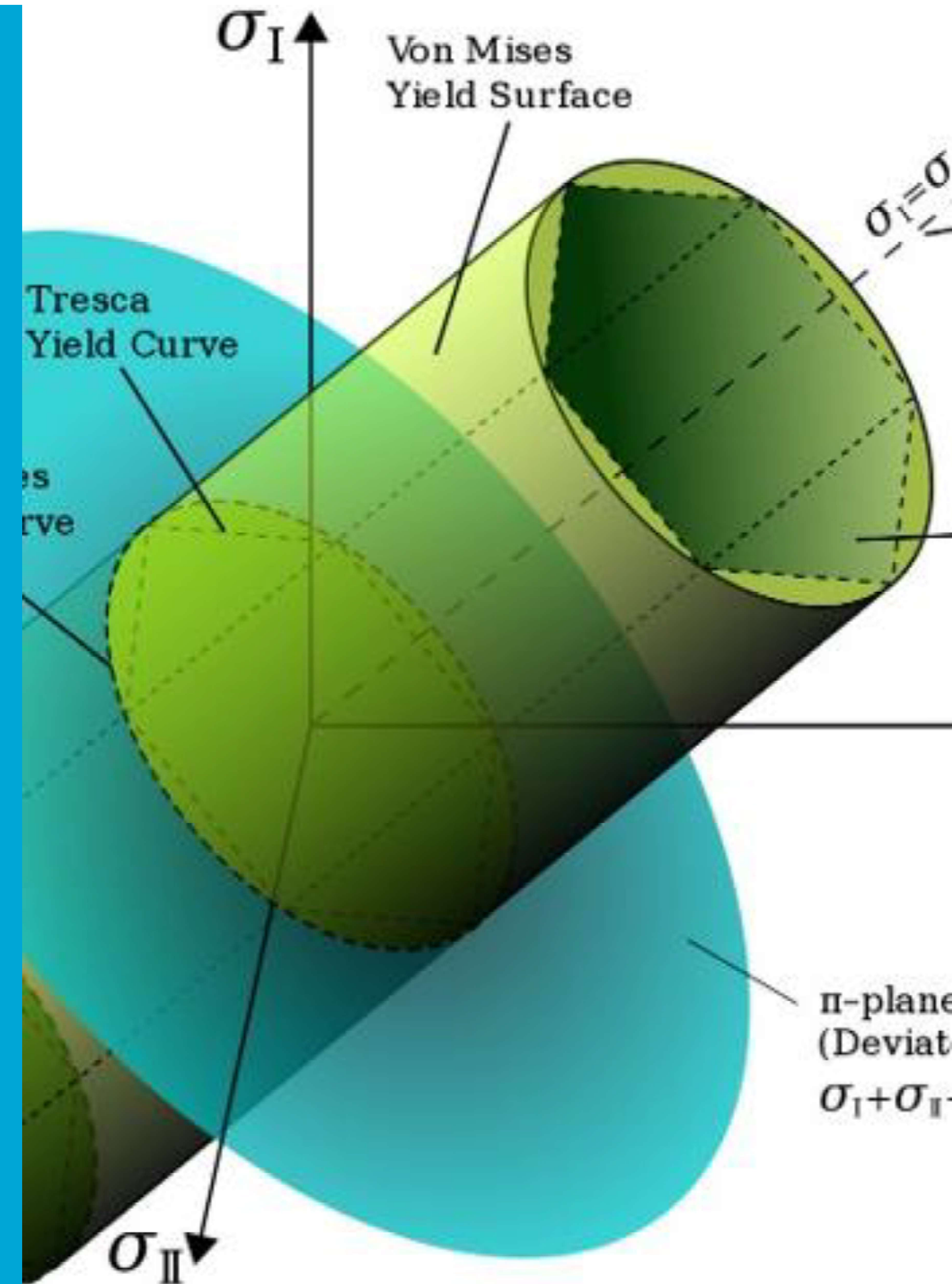


Theory of elasticity

1, 2, and 7 September 2026

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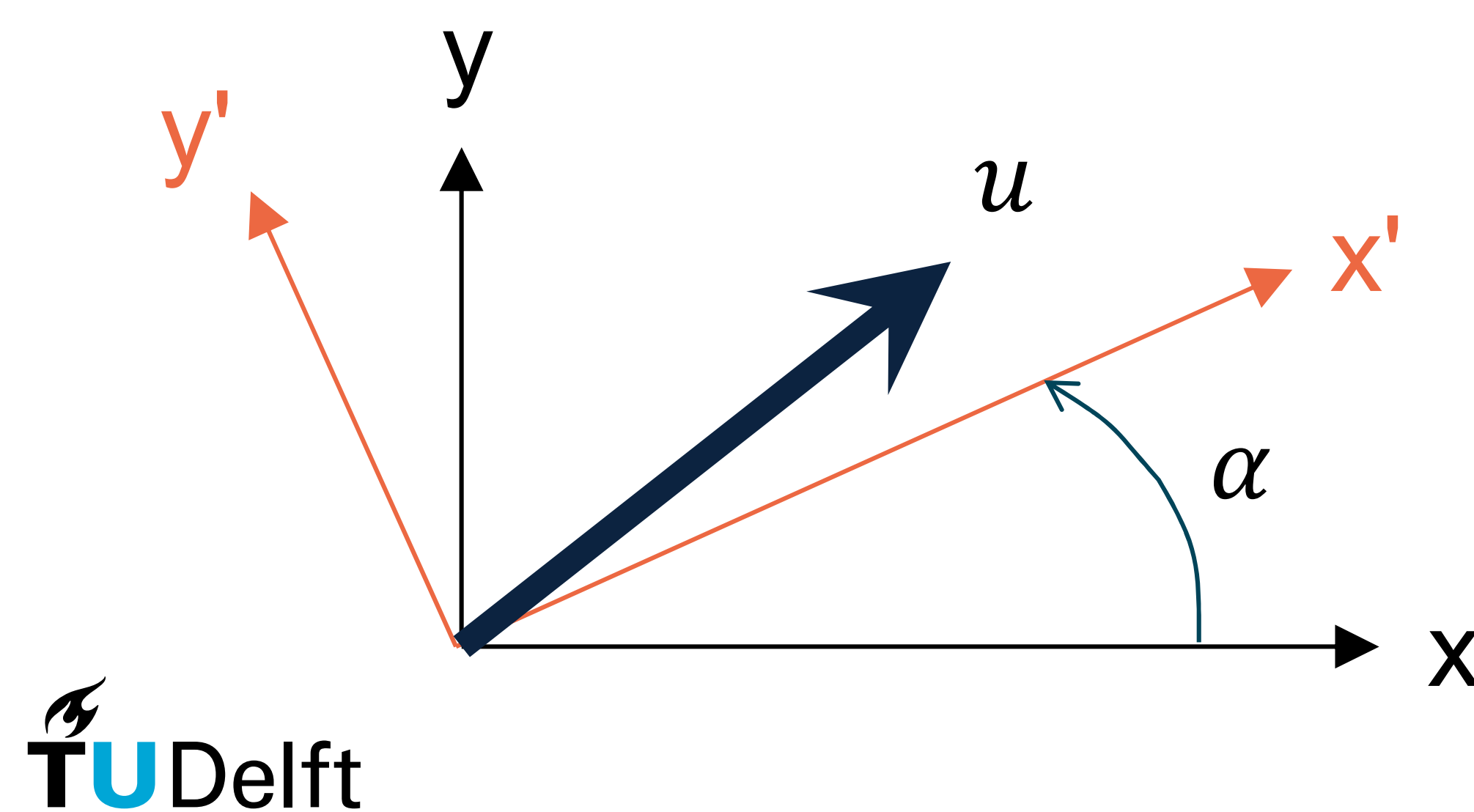


02

Tensor transformations and principal values

First order tensor transformation

- A **first order tensor** can be seen as **a vector with a magnitude**
- The magnitude is constant and does not vary depending on the coordinate system, but the direction depends on the coordinate system
- When the coordinate system changes, the tensor changes:



Transformation matrix R

$$u' = Ru \quad \text{or} \quad \begin{bmatrix} u_{x'} \\ u_{y'} \end{bmatrix} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} u_x \\ u_y \end{bmatrix}$$

Second order tensor transformation

- A **second order tensor** too follows a transformation rule like that of the 1st order tensors:

$$[\sigma'] = [R][\sigma][R^T] \quad \text{or} \quad \begin{bmatrix} \sigma_{x'x'} & \sigma_{x'y'} \\ \sigma_{x'y'} & \sigma_{y'y'} \end{bmatrix} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{yy} \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

- This rule can be expressed as:

$$\sigma'_{xx} = \sigma_{x'x'} = \frac{1}{2}(\sigma_{xx} + \sigma_{yy}) + \frac{1}{2}(\sigma_{xx} - \sigma_{yy}) \cos 2\alpha + \sigma_{xy} \sin 2\alpha$$

$$\sigma'_{yy} = \sigma_{y'y'} = \frac{1}{2}(\sigma_{xx} + \sigma_{yy}) - \frac{1}{2}(\sigma_{xx} - \sigma_{yy}) \cos 2\alpha - \sigma_{xy} \sin 2\alpha$$

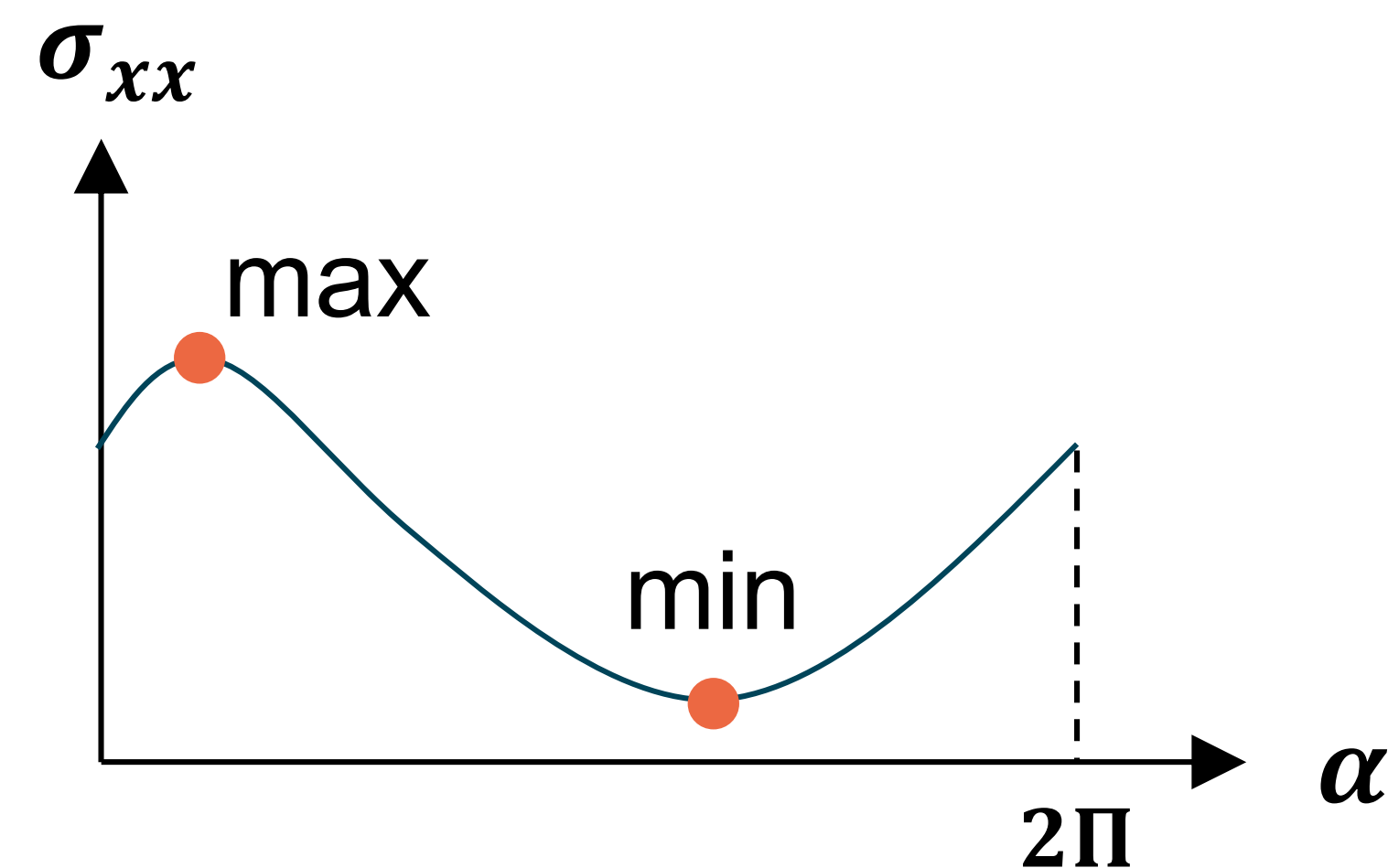
$$\sigma'_{xy} = \sigma_{x'y'} = -\frac{1}{2}(\sigma_{xx} - \sigma_{yy}) \sin 2\alpha + \sigma_{xy} \cos 2\alpha$$

*the problem is presented here for stresses, but the transformation rules apply equally to strains (and in general to every 2nd order tensor)

Principal stresses (and strains*)

We want to find a direction for which the shear stress is zero!

$$\boldsymbol{\sigma}^{(1)} = \mathbf{R}\boldsymbol{\sigma}\mathbf{R}^T \quad \text{or} \quad \begin{bmatrix} \sigma_{xx}^{(1)} & \sigma_{xy}^{(1)} \\ \sigma_{xy}^{(1)} & \sigma_{yy}^{(1)} \end{bmatrix} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{yy} \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$



The diagonal terms $(\sigma_{xx}^{(1)}, \sigma_{yy}^{(1)})$ are maximised or minimised when its derivative with respect to the angle α is zero:

$$\frac{d\sigma_{xx}^{(1)}}{d\alpha} = 0 \quad \text{and} \quad \frac{d\sigma_{yy}^{(1)}}{d\alpha} = 0$$

*the problem is defined here for stresses, but each step is equivalent for strains (and in general for every 2nd order tensor)

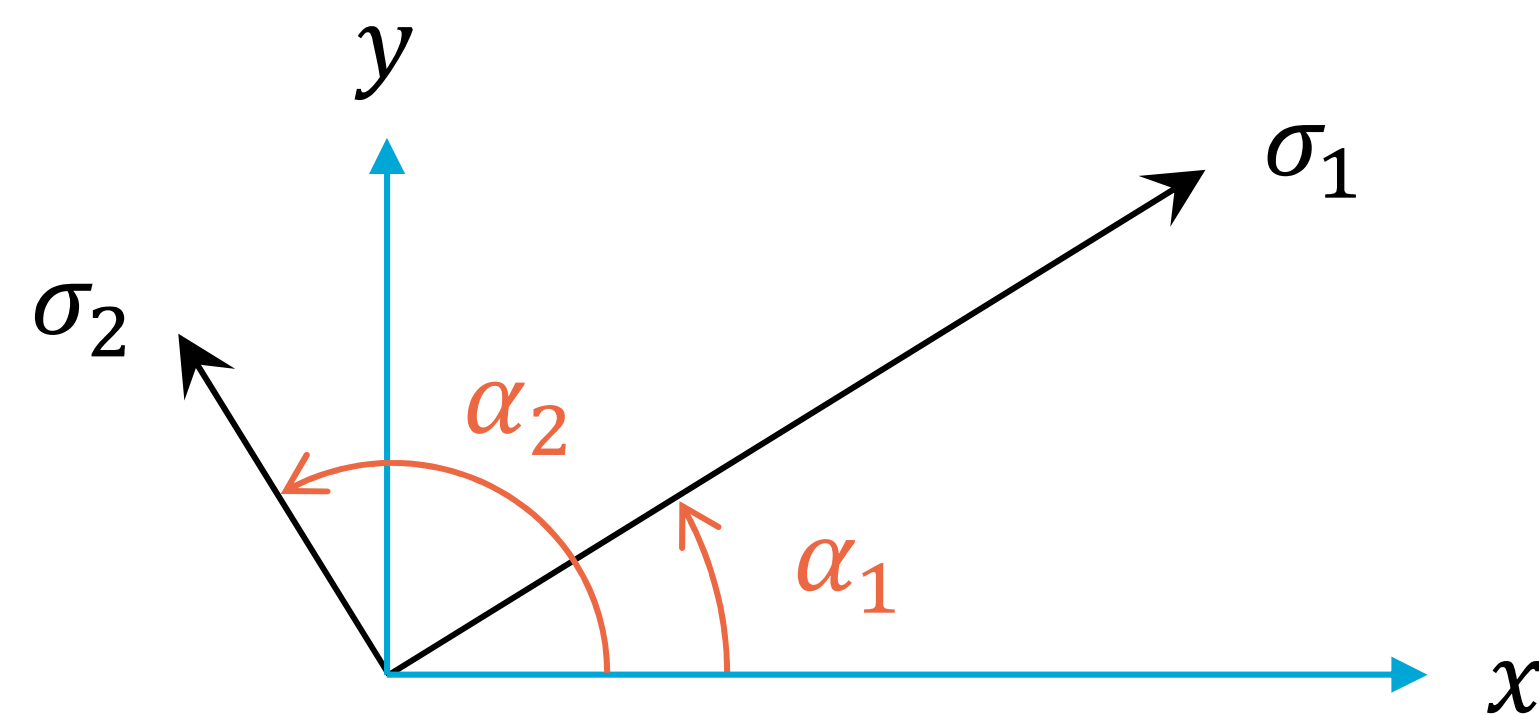
Principal stresses (and strains)

We obtain:

Equivalent to solution of an eigenvalue problem

$$\tan 2\alpha_1 = \frac{2\sigma_{xy}}{\sigma_{xx} - \sigma_{yy}}, \quad \alpha_2 = \alpha_1 + 90^\circ \text{ and } \sigma_{1;2} = \frac{1}{2}(\sigma_{xx} + \sigma_{yy}) \pm \sqrt{\left[\frac{1}{2}(\sigma_{xx} - \sigma_{yy})\right]^2 + \sigma_{xy}^2}$$

$\sigma_{1;2}$ are the extreme of the stress tensor, and are called **principal stresses** (being σ_1 the maximum principal stress and σ_2 the minimum principal stress)



Since the eigenvectors (i.e. the solutions of an eigenvalue problem) are orthogonal one to another, the non-diagonal terms of the transformed matrix by an angle α_1 become equal to 0:

$$\sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}$$

Symmetric 2nd order tensors as vectors

Symmetric second order tensors can be mapped in a vector form:

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} & \sigma_{yx} & \sigma_{zx} \\ \sigma_{xy} & \sigma_{yy} & \sigma_{zy} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz} \end{bmatrix} \Rightarrow \sigma = \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{xz} \\ \sigma_{yz} \end{bmatrix}$$

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{yx} & \varepsilon_{zx} \\ \varepsilon_{xy} & \varepsilon_{yy} & \varepsilon_{zy} \\ \varepsilon_{xz} & \varepsilon_{yz} & \varepsilon_{zz} \end{bmatrix} \Rightarrow \varepsilon = \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ 2\varepsilon_{xy} \\ 2\varepsilon_{xz} \\ 2\varepsilon_{yz} \end{bmatrix} = \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{xz} \\ \gamma_{yz} \end{bmatrix}$$

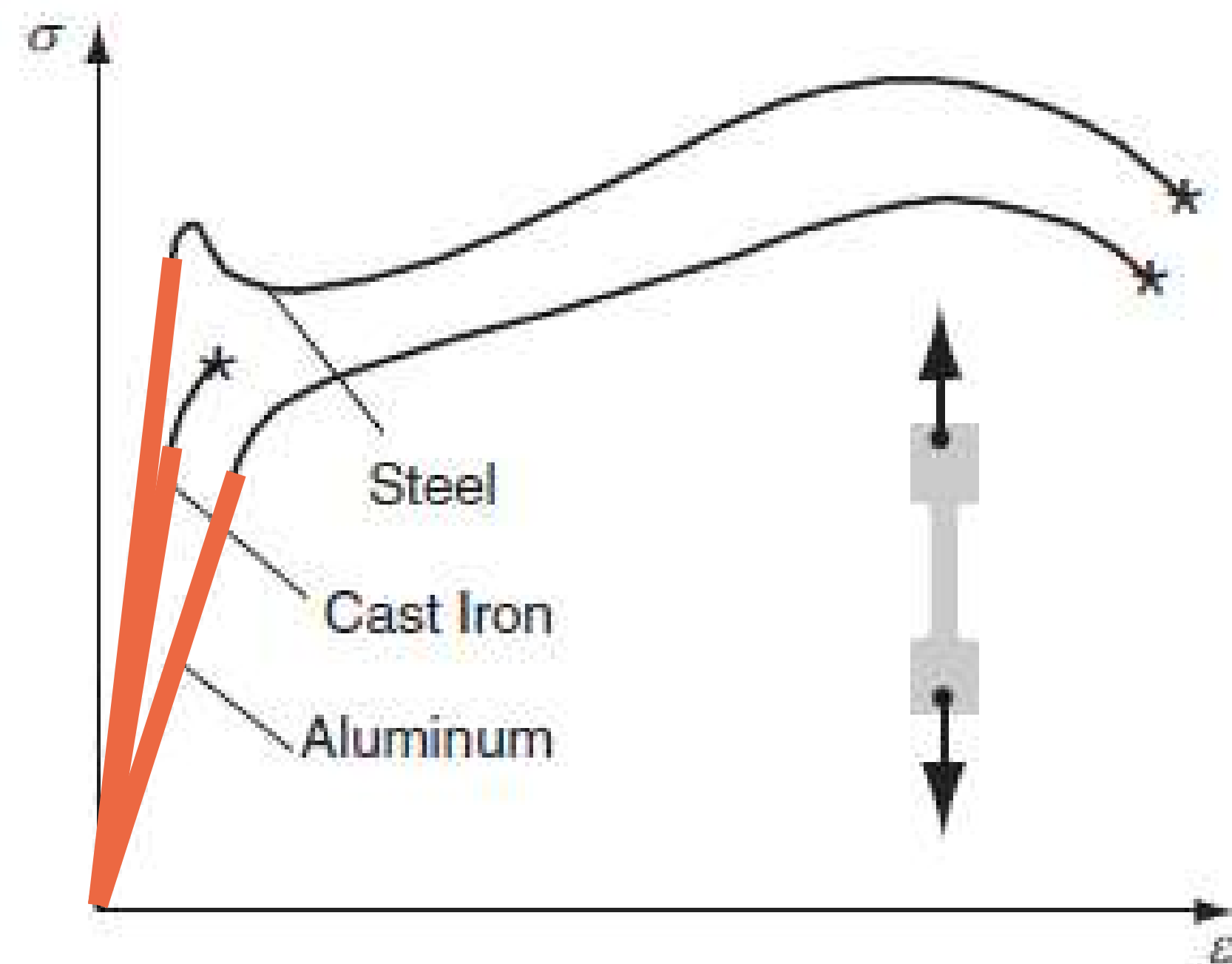
Required to ensure energy equivalence
(for strain energy density $U = \frac{1}{2} \boldsymbol{\varepsilon} : \boldsymbol{\sigma}$ - CM4)

03

Stress-strain relation for linear elasticity

Constitutive models

- Relation between stress and strains = **constitutive model**
- Constitutive model depends on the material
- Simplest case/approximation: **linear relation** → valid up to the **elastic limit**



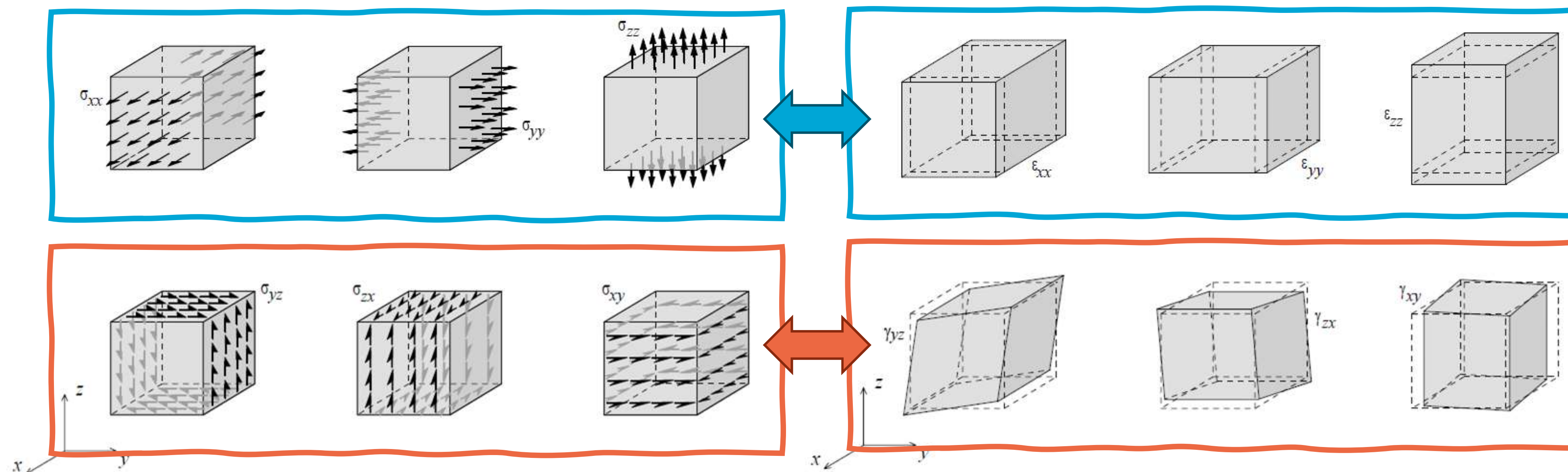
$$\boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{xy} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{xz} & \epsilon_{yz} & \epsilon_{zz} \end{bmatrix} \xleftrightarrow[\text{?}]{\text{Linear relation}} \boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz} \end{bmatrix}$$

$$\boldsymbol{\sigma} = \mathbb{C} \boldsymbol{\epsilon}$$

Linear elastic isotropic materials

For linear elastic isotropic materials:

- only **2 independent parameters**: $\boldsymbol{\sigma} = \mathbb{C}\boldsymbol{\varepsilon} = \mathbb{C}(E, \nu)\boldsymbol{\varepsilon}$
- Deformations due to normal stresses are uncoupled from shear deformations



$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \end{bmatrix} = \mathbb{C}^{(1)}(E, \nu) \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \end{bmatrix}$$

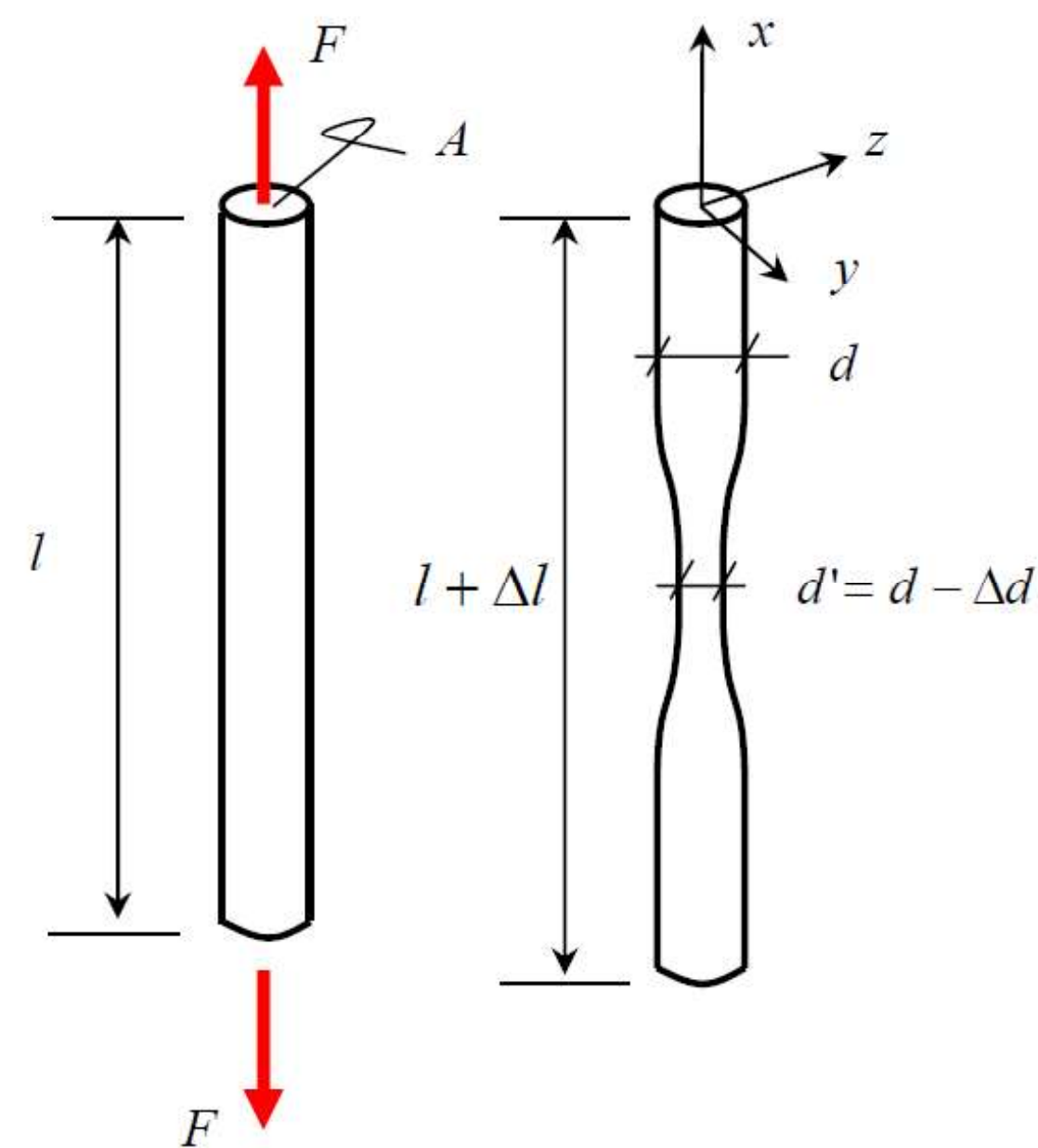
+

$$\begin{bmatrix} \sigma_{xy} \\ \sigma_{xz} \\ \sigma_{yz} \end{bmatrix} = \mathbb{C}^{(2)}(E, \nu) \begin{bmatrix} \varepsilon_{xy} \\ \varepsilon_{xz} \\ \varepsilon_{yz} \end{bmatrix}$$

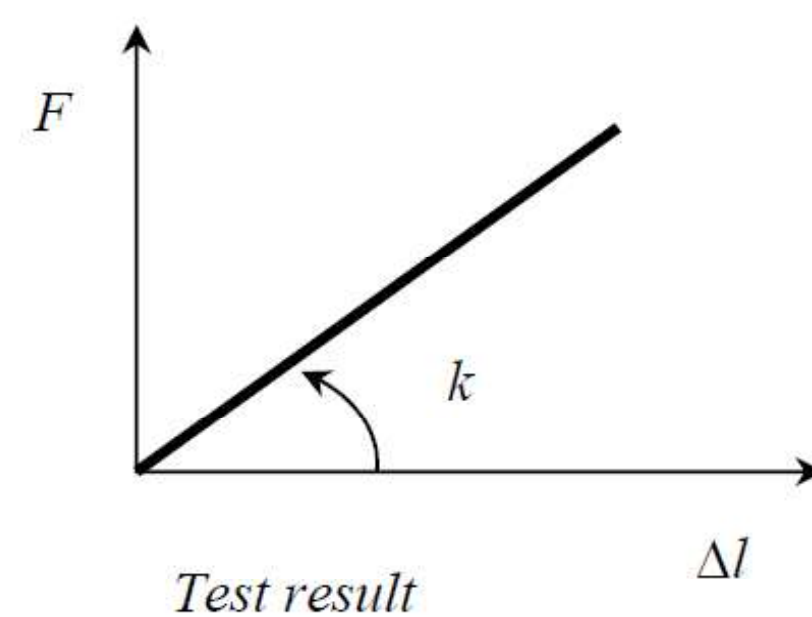
$$\mathbb{C} = \begin{bmatrix} \mathbb{C}^{(1)} & \mathbf{0} \\ \mathbf{0} & \mathbb{C}^{(2)} \end{bmatrix}$$

Linear elasticity: normal stresses vs strains

Uniaxial test



Hooke's law



Elongation: $\epsilon_{xx} = \frac{\sigma_{xx}}{E}$

Necking: $\epsilon_{yy} = \epsilon_{zz} = -\nu \frac{\sigma_{xx}}{E}$

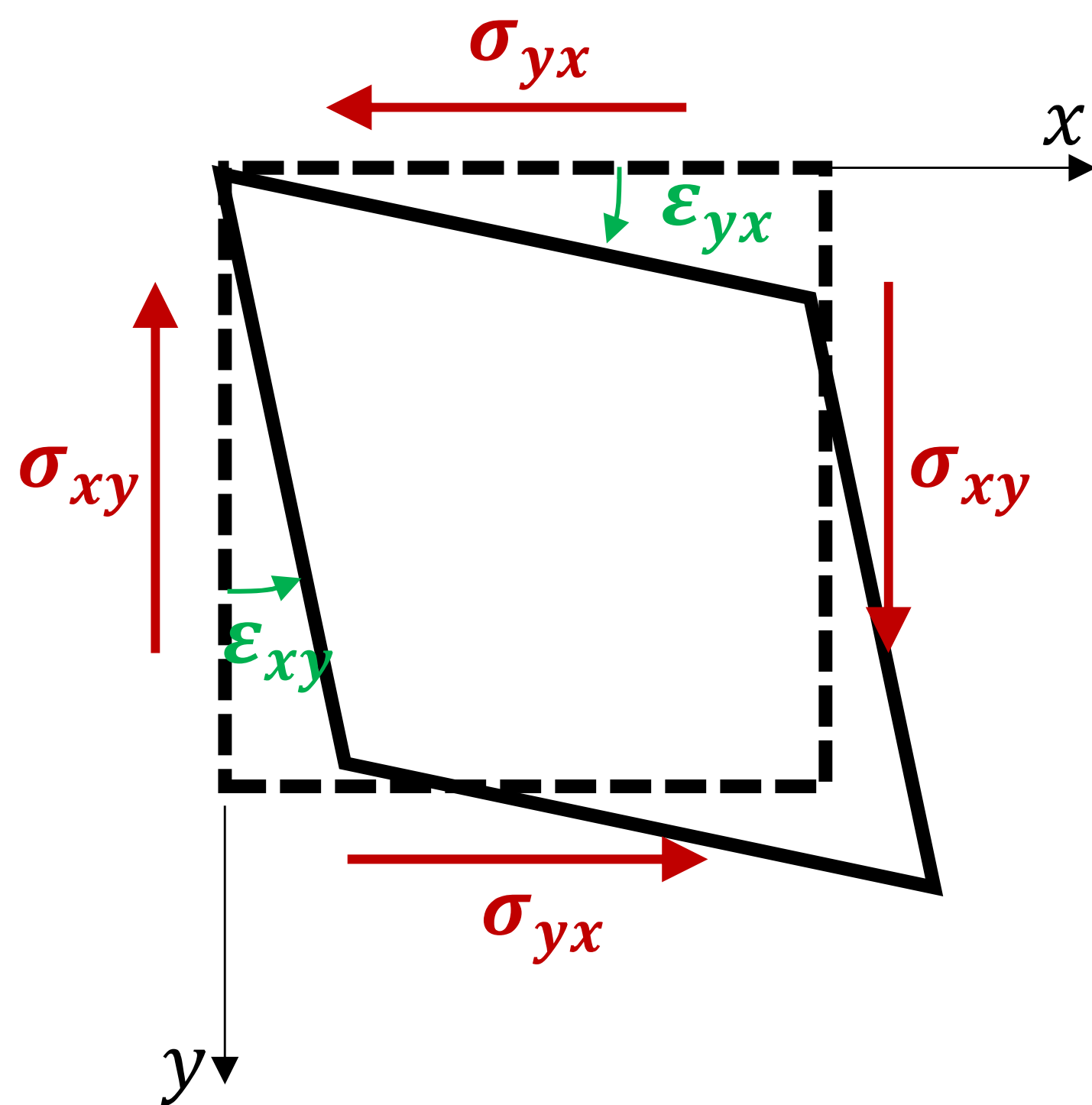
General 3D case

Principle of superposition:

- $\epsilon_{xx} = \frac{\sigma_{xx}}{E} - \frac{\nu\sigma_{yy}}{E} - \frac{\nu\sigma_{zz}}{E}$
- $\epsilon_{yy} = -\frac{\nu\sigma_{xx}}{E} + \frac{\sigma_{yy}}{E} - \frac{\nu\sigma_{zz}}{E}$
- $\epsilon_{zz} = -\frac{\nu\sigma_{xx}}{E} - \frac{\nu\sigma_{yy}}{E} + \frac{\sigma_{zz}}{E}$

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu \\ -\nu & 1 & -\nu \\ -\nu & -\nu & 1 \end{bmatrix} \begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{zz} \end{bmatrix}$$

Linear elasticity: shear stresses vs strains



- We define the **shear modulus G**:

$$G = \frac{\sigma_{xy}}{\gamma_{xy}}$$

- **G** is a **material property** and holds for all directions:

$$G = \frac{\sigma_{xy}}{\gamma_{xy}} = \frac{\sigma_{xz}}{\gamma_{xz}} = \frac{\sigma_{yz}}{\gamma_{yz}}$$

Since $\gamma_{ij} = 2\varepsilon_{ij}$: **$\sigma_{ij} = 2G\varepsilon_{ij}$ or $\varepsilon_{ij} = \frac{\sigma_{ij}}{2G}$ ($i, j = x, y, z$)**

For a linear elastic material, G is not an independent property: **$G = \frac{E}{2(1+\nu)}$**

Constitutive law: 3D strain-stress relation

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \mathbf{2}\varepsilon_{xy} \\ \mathbf{2}\varepsilon_{yz} \\ \mathbf{2}\varepsilon_{zx} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu & 0 & 0 & 0 \\ -\nu & 1 & -\nu & 0 & 0 & 0 \\ -\nu & -\nu & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2(1+\nu) & 0 & 0 \\ 0 & 0 & 0 & 0 & 2(1+\nu) & 0 \\ 0 & 0 & 0 & 0 & 0 & 2(1+\nu) \end{bmatrix} \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{yz} \\ \sigma_{zx} \end{bmatrix}$$

Constitutive law: 3D stress-strain relation

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{yz} \\ \sigma_{zx} \end{bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ 2\varepsilon_{xy} \\ 2\varepsilon_{yz} \\ 2\varepsilon_{zx} \end{bmatrix}$$

Constitutive law: plane stress conditions

- Stresses in z direction are negligible ($\sigma_{zz}, \sigma_{xz}, \sigma_{yz} \approx 0$)

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{xy} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & 0 \\ -\nu & 1 & 0 \\ 0 & 0 & 1 + \nu \end{bmatrix} \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{bmatrix}$$

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{bmatrix} = \frac{E}{1 - \nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & 1 - \nu \end{bmatrix} \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{xy} \end{bmatrix}$$



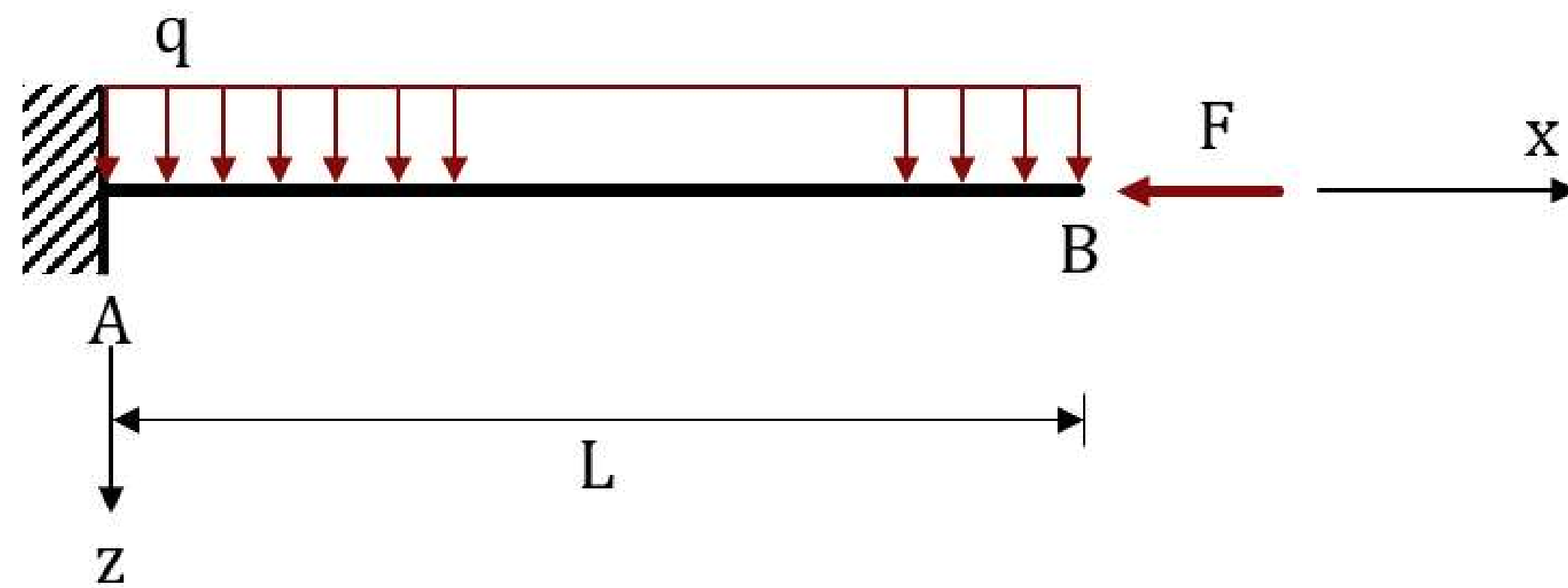
Constitutive law: principal stresses and strains

- Principal stresses and strains can be described by a first order tensor (vector) with the diagonal terms of the second order tensor (matrix)
- All the shear stresses and strains are zero, leading to simplified expressions:

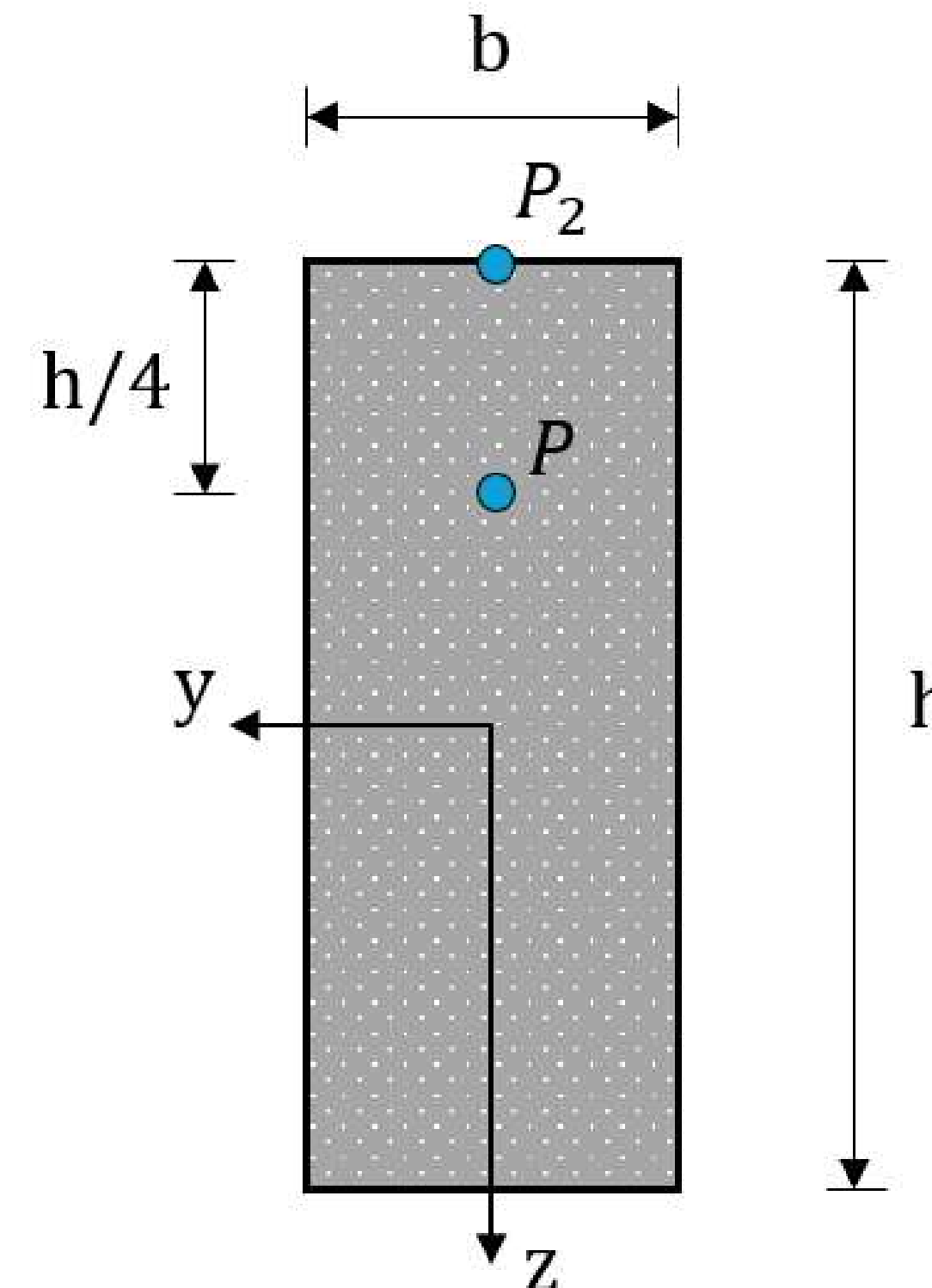
- $\varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu \\ -\nu & 1 & -\nu \\ -\nu & -\nu & 1 \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \end{bmatrix}$ and $\varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu \\ -\nu & 1 \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix}$ for plane stress conditions

- $\sigma = \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \end{bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & \nu \\ \nu & 1 & \nu \\ \nu & \nu & 1 \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix}$ and $\sigma = \begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \end{bmatrix}$ for plane stress conditions

Example (#02 – part A)



Rectangular section:



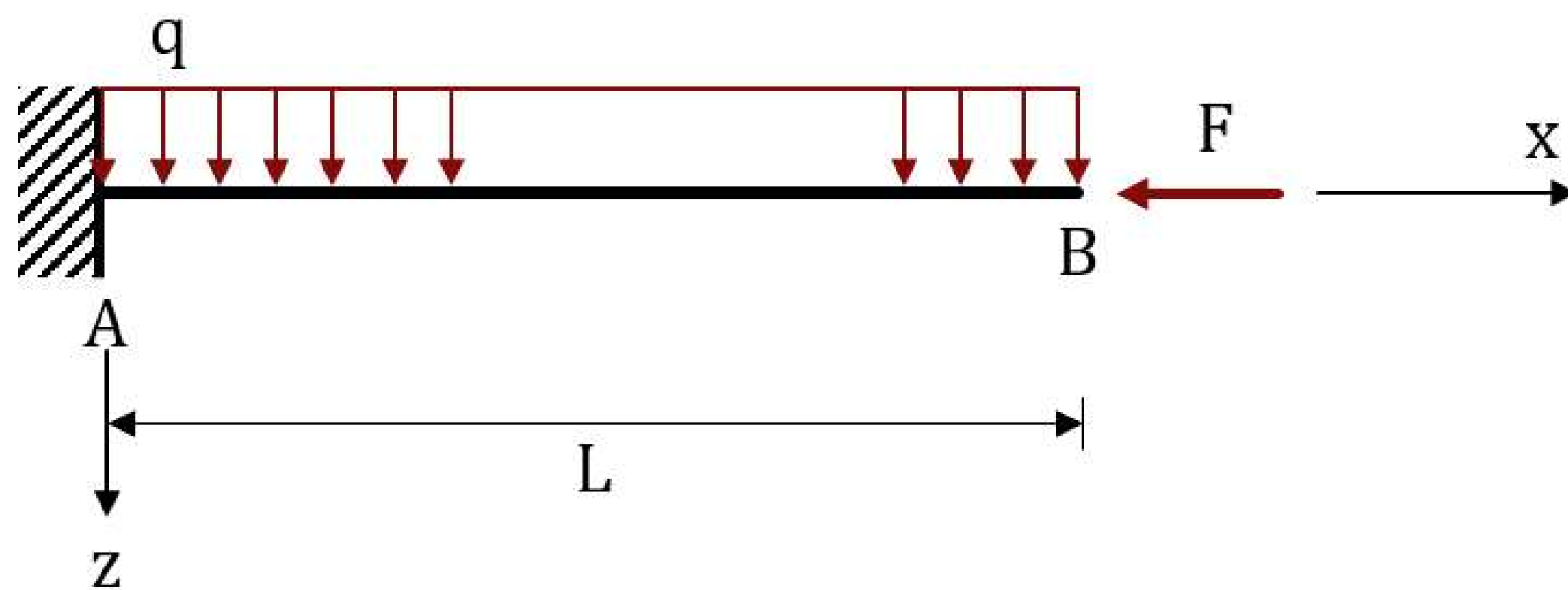
$$\begin{aligned} F &= -48 \text{ kN} \\ q &= 12 \text{ kN/m} \\ \nu &= 0.2 \\ E &= 30'000 \text{ MPa} \\ L &= 2 \text{ m} \end{aligned}$$

Rectangular
section:
 $b = 200 \text{ mm}$
 $h = 600 \text{ mm}$

Considering plane-stress conditions, compute at the point P of section A:

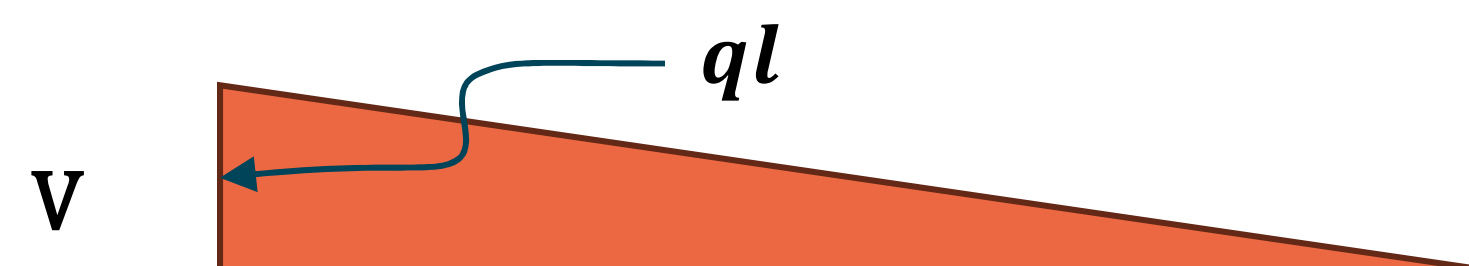
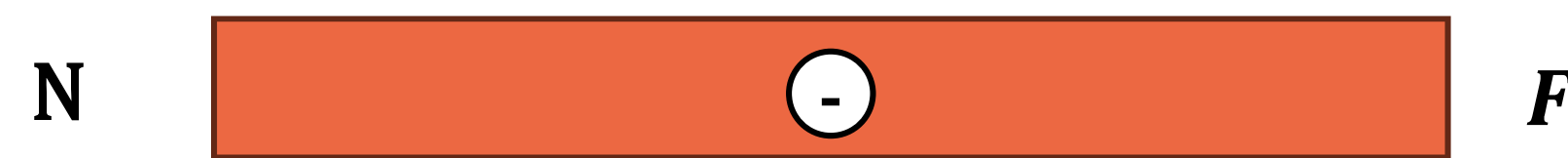
1. The stress tensor
2. Principal stresses
3. Principal strains

Example (#02 – part A) - solution procedure



We first compute the internal forces at section A, for which we have:

- $N_A = F = -48 \text{ kN}$
- $V_A = qL = 24 \text{ kN}$
- $M_A = -\frac{qL^2}{2} = -24 \text{ kNm}$



Example (#02 – part A) - solution procedure

We compute then the stresses at point P:

- Given the plane-stress condition:

$$\sigma_{yy} = \sigma_{xy} = \sigma_{yz} = \sigma_{zz} = 0$$

- The normal stress is:

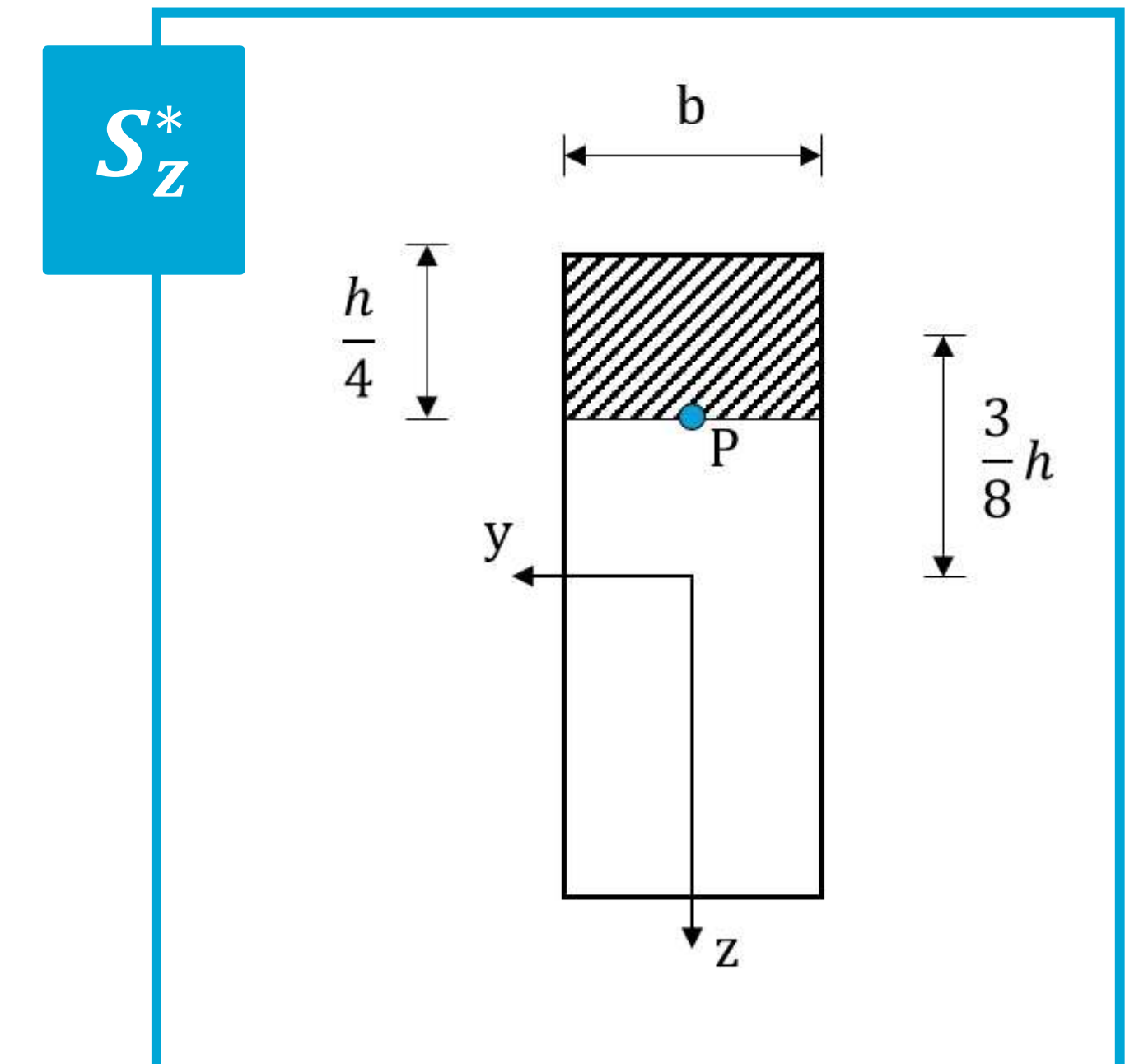
$$\sigma_{xx} = \frac{M_A z_P}{I_{zz}} + \frac{N_A}{A} = \frac{M_A(-h/4)}{I_{zz}} + \frac{N_A}{A} = 0.6 \text{ MPa}$$

- The shear stress is:

$$\sigma_{xz} = \frac{V_A S_z^*(z_P)}{I_{zz} b} = 0.225 \text{ MPa}$$

being:

- $A = bh = 1.2 \cdot 10^5 \text{ mm}^2$
- $I_{zz} = \frac{bh^3}{12} = 3.6 \cdot 10^9 \text{ mm}^4$
- $S_z^*(z_P) = b \cdot \frac{h}{4} \cdot \frac{3}{8}h = \frac{3bh^2}{32} = 6.75 \cdot 10^6 \text{ mm}^3$



Example (#02 – part A) - solution procedure

1. The stress tensor in xyz-coordinates is: $\sigma = \begin{bmatrix} 0.6 & 0 & 0.225 \\ 0 & 0 & 0 \\ 0.225 & 0 & 0 \end{bmatrix} MPa$
2. Since it is a plane-stress condition ($\sigma_{yy} = \tau_{xy} = \tau_{yz} = 0$) we can compute the principal stresses as:

$$\sigma_1 = \frac{1}{2}(\sigma_{xx} + \sigma_{zz}) + \sqrt{\left[\frac{1}{2}(\sigma_{xx} - \sigma_{zz})\right]^2 + \sigma_{xz}^2} = \frac{\sigma_{xx}}{2} + \sqrt{\left(\frac{\sigma_{xx}}{2}\right)^2 + \sigma_{xz}^2} = 0.675 MPa$$

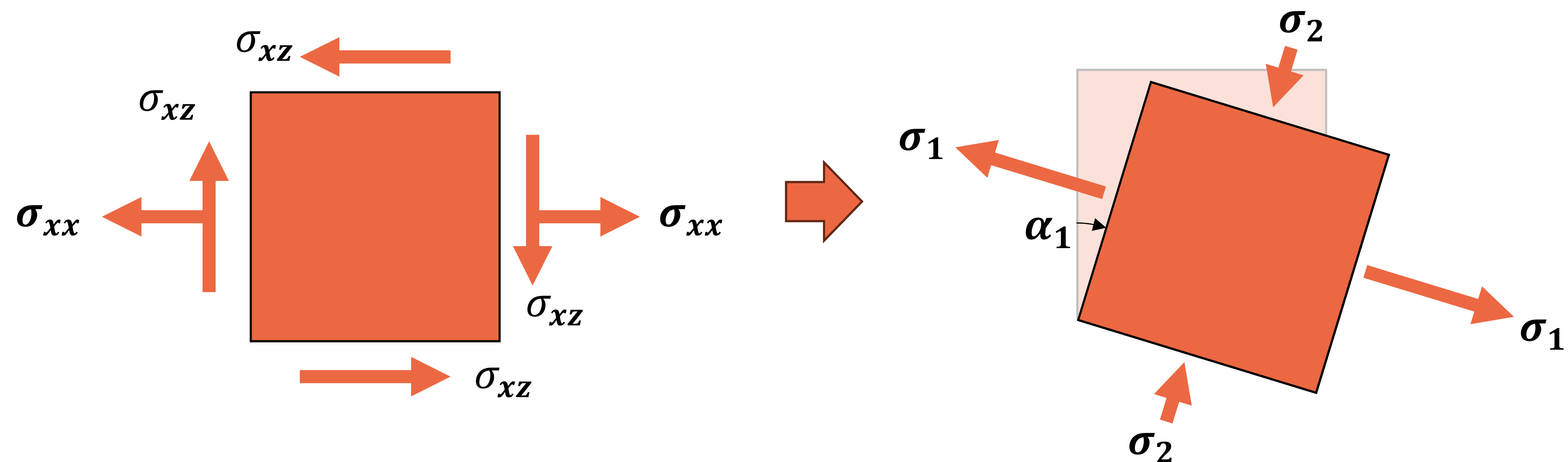
$$\sigma_2 = \frac{1}{2}(\sigma_{xx} + \sigma_{zz}) - \sqrt{\left[\frac{1}{2}(\sigma_{xx} - \sigma_{zz})\right]^2 + \sigma_{xz}^2} = \frac{\sigma_{xx}}{2} - \sqrt{\left(\frac{\sigma_{xx}}{2}\right)^2 + \sigma_{xz}^2} = -0.075 MPa$$

Example (#02 – part A) - solution procedure

- To compute the direction of the principal stresses we refer to the equation:

$$\tan 2\alpha_1 = \frac{2\sigma_{xz}}{\sigma_{xx} - \sigma_{zz}} = \frac{2\sigma_{xz}}{\sigma_{xx}} = 0.75$$

- The corresponding angle is: $\alpha_1 = 0.32 \text{ rad} = 18.4^\circ$, while $\alpha_2 = \alpha_1 - 90^\circ = -71.6^\circ$



Example (#02 – part A) - solution procedure

3. We start from the computed principal stresses:

$$\sigma = \begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix} = \begin{bmatrix} 0.675 \\ -0.075 \end{bmatrix} MPa$$

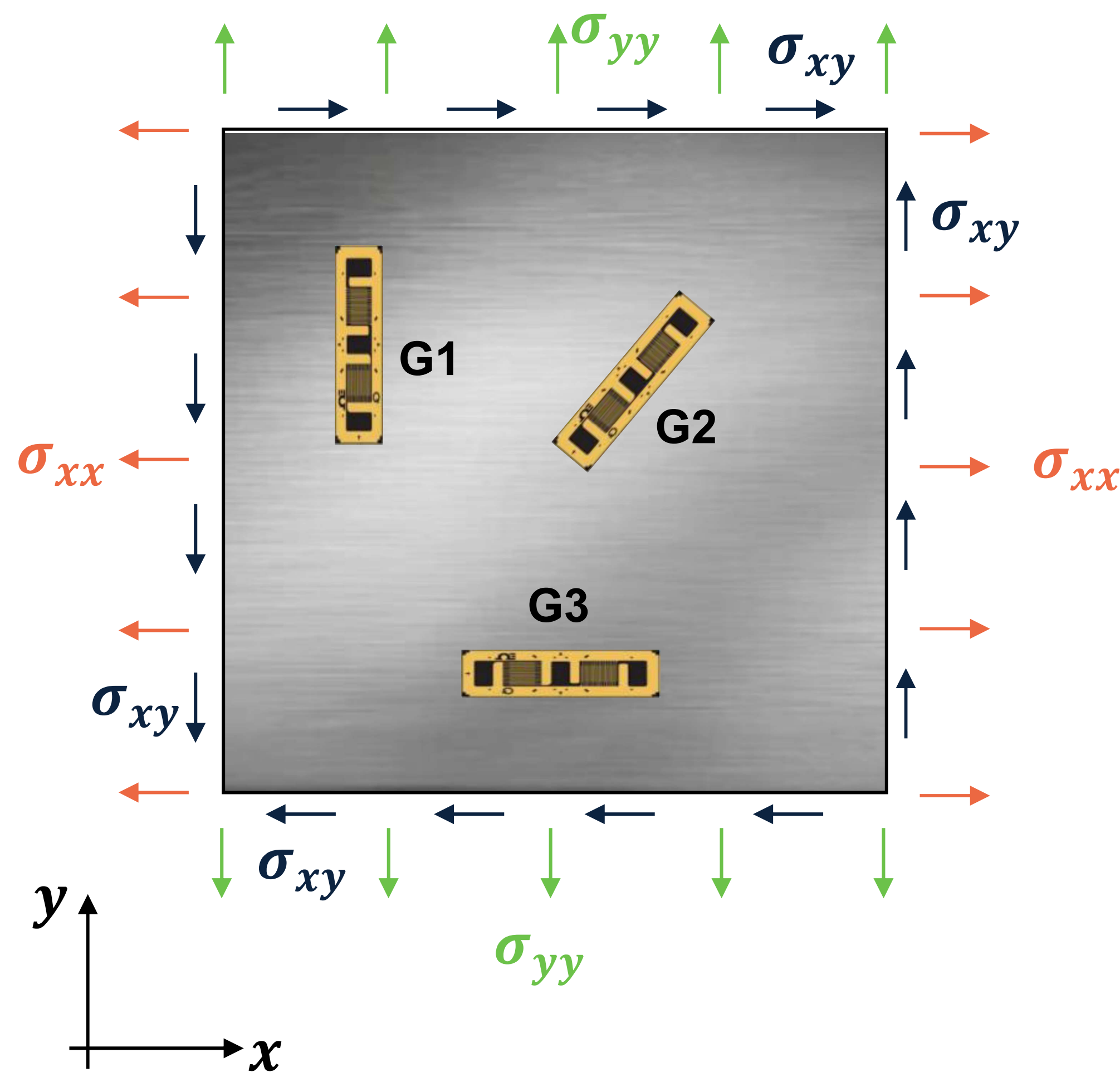
- We apply the constitutive law for principal stresses and strains:

$$\varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu \\ -\nu & 1 \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix} = \frac{1}{30000} \begin{bmatrix} 1 & -0.2 \\ -0.2 & 1 \end{bmatrix} \begin{bmatrix} 0.675 \\ -0.075 \end{bmatrix} = \begin{bmatrix} 2.3 \\ -0.7 \end{bmatrix} 10^{-5}$$

- The principal strains have the same orientation of the principal stresses, so they are rotated by an angle $\alpha = 18.4^\circ$ with respect to the xy-coordinates

Example (#03)

*strain gauges read deformations only along their axis



A plate of aluminium ($E = 70000 \text{ MPa}$, $\nu = 0.33$, $f_y = 300 \text{ MPa}$) is tested in **homogeneous plane stress conditions**.

Three **strain gauges*** are mounted on the specimen:

- $G1$: oriented along the y -axis - $\varepsilon_{G1} = -1.0 \cdot 10^{-3}$
- $G2$: with an inclination of 45° to the x -axis - $\varepsilon_{G2} = 2.0 \cdot 10^{-3}$
- $G3$: oriented along the x -axis - $\varepsilon_{G3} = 2.0 \cdot 10^{-3}$

Compute:

1. The strain tensor
2. the principal strains
3. The principal stresses

Example (#03) - solution procedure

- Based on the transformation rules, we consider the strains aligned with two coordinate systems: xy-coordinates and x'y'-coordinates, rotated by 45° one to another

- The normal strain aligned with the first axis is then:

$$\varepsilon_{45} = \frac{1}{2}(\varepsilon_{xx} + \varepsilon_{yy}) + \frac{1}{2}(\varepsilon_{xx} - \varepsilon_{yy}) \cos 2\alpha + \varepsilon_{xy} \sin 2\alpha$$

- We can replace the known information and we obtain:

$$\varepsilon_{G2} = \frac{1}{2}(\varepsilon_{G3} + \varepsilon_{G1}) + \frac{1}{2}(\varepsilon_{G3} - \varepsilon_{G1}) \cos 2(45^\circ) + \varepsilon_{xy} \sin 2(45^\circ)$$

- The only unknown of this expression is ε_{xy} and we can therefore compute it (see next slide)

Example (#03) - solution procedure

$$\varepsilon_{xy} = \frac{\varepsilon_{G2} - \frac{1}{2}(\varepsilon_{G3} + \varepsilon_{G1}) - \frac{1}{2}(\varepsilon_{G3} - \varepsilon_{G1}) \cos(90^\circ)}{\sin(90^\circ)} = \varepsilon_{G2} - \frac{1}{2}(\varepsilon_{G3} + \varepsilon_{G1}) = 1.5 \cdot 10^{-3}$$

- We can now write the strain tensor:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} 2 & 1.5 \\ 1.5 & -1 \end{bmatrix} 10^{-3}$$

- The principal strains are:

$$\varepsilon_1 = \frac{1}{2}(\varepsilon_{xx} + \varepsilon_{yy}) + \sqrt{\left[\frac{1}{2}(\varepsilon_{xx} - \varepsilon_{yy})\right]^2 + \varepsilon_{xy}^2} = 2.62 \cdot 10^{-3}$$

$$\varepsilon_2 = \frac{1}{2}(\varepsilon_{xx} + \varepsilon_{yy}) - \sqrt{\left[\frac{1}{2}(\varepsilon_{xx} - \varepsilon_{yy})\right]^2 + \varepsilon_{xy}^2} = -1.62 \cdot 10^{-3}$$

Example (#03) - solution procedure

- Since it is requested to compute the stress tensor in the principal directions, we can apply the constitutive law for plane stress conditions, using the strain tensor previously computed

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix} = \frac{E}{1 - \nu^2} \begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \end{bmatrix} = \begin{bmatrix} 164 \\ -59.4 \end{bmatrix} \text{MPa}$$

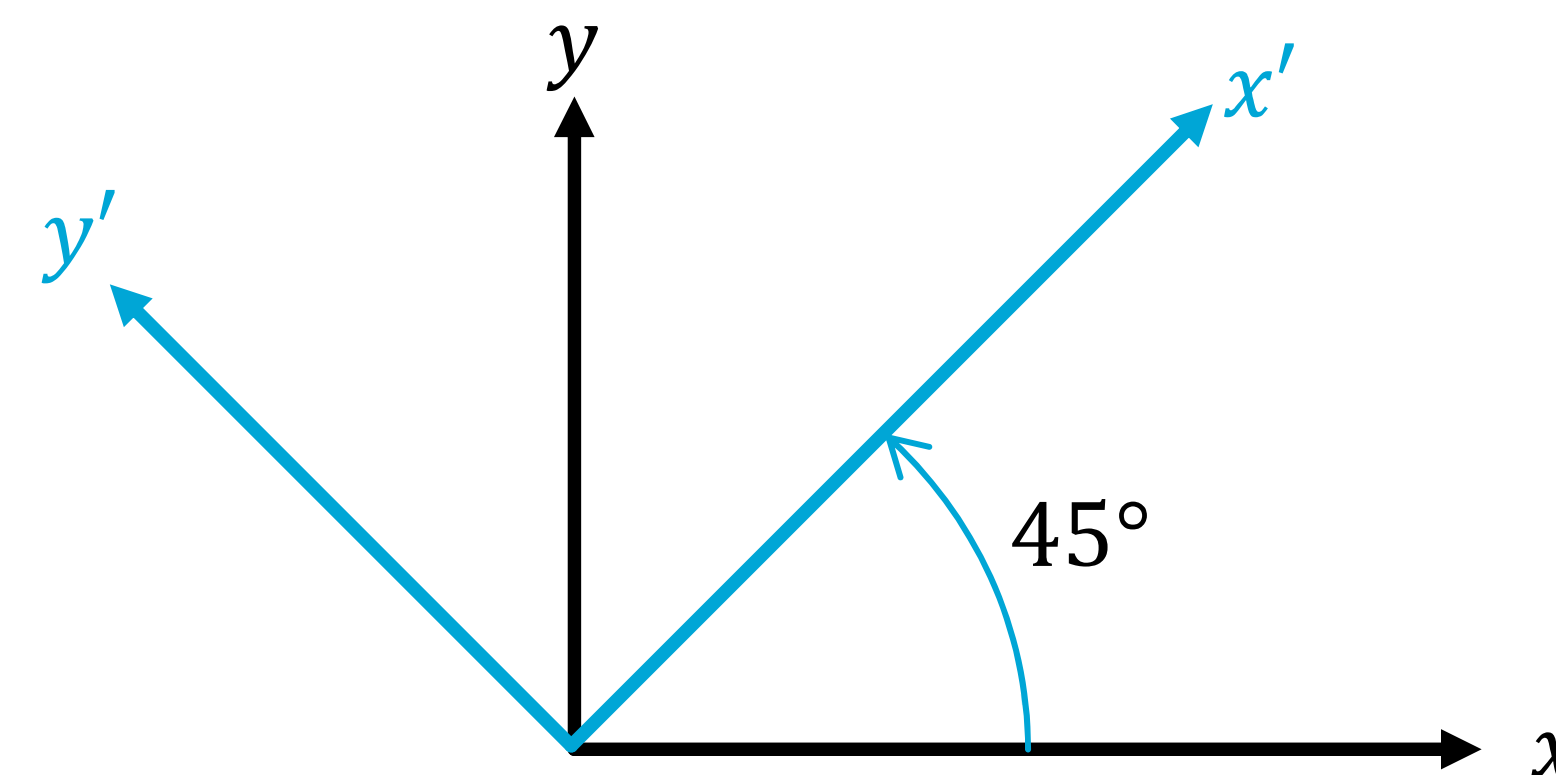
Guided exercise #02

In a steel beam ($E = 210'000 \text{ MPa}$, $\nu = 0.2$), three strain gauges measure at point P the strains along an angle inclined of 45° with respect to the xy-axes:

$$[\varepsilon'] = \begin{bmatrix} 5.0 & 1.0 \\ 1.0 & -2.0 \end{bmatrix} \cdot 10^{-4}$$

Assuming plane stress conditions, compute at point P:

- the stress tensor in xy-coordinates;
- the principal stresses.



Guided exercise #03

At point P of a concrete beam ($E = 30'000 \text{ MPa}$, $\nu = 0.3$), the following stress status is estimated:

$$[\sigma] = \begin{bmatrix} 120 & -45 \\ -45 & 0 \end{bmatrix} \text{MPa}$$

Compute the:

- the strain tensor in xy-coordinates;
- the principal strains.